Hyperliquid Hourly Funding Rate, Open Interest Crowding & Squeeze Regime Prediction

Quantitative Research Brief: Hourly Funding Rate & Open Interest Crowding Integration for HMM Regime Switching & O-U Drift on Hyperliquid

1. System Context & Production Architecture

We run an autonomous Rust trading bot on Hyperliquid Perpetuals (`BTC`, `ETH`, `SOL`, `AVAX`) driven by a 3-state, 4-feature Gaussian Hidden Markov Model (`GaussianHMM` with Ledoit-Wolf variance shrinkage and semantic state sorting: `State 0 = Consolidation`, `State 1 = Trend`, `State 2 = Volatility Shock`).

- Currently, the 4 HMM features (computed on 15m bars) are strictly price-based:
  1. Volatility-normalized return r_t / σ_t
  2. Log realized volatility $\ln(\sigma_t)$
  3. Kaufman Efficiency Ratio $\mathrm{KER}_{16}(t)$
  4. Absolute directional drift $|D_{16}(t)|$
- We also stream Hyperliquid's `ActiveAssetCtx` WebSocket channel (`PerpsAssetCtx`), which provides real-time **hourly** `funding`, `open_interest`, `oracle_px`, and `mark_px` (`mark - oracle` basis premium).
- On Hyperliquid, funding is settled **every hour** at $1/8\text{th}$ of the 8-hour rate based on the mark-oracle premium index plus a clamped interest rate component.

2. Core Research Questions to Investigate

1. **Distinguishing Spot-Driven Healthy Trends from Leveraged Blow-Off Squeezes:**
   - Price-only HMMs cannot distinguish between:
     - **Regime T1 (Healthy Spot-Led Trend):** Price rises with moderate/neutral funding and steady Open Interest (OI) absorption $\rightarrow$ high probability of trend continuation (ideal for Engine B Donchian breakout).
     - **Regime T2 (Crowded Perp-Led Exhaustion / Squeeze Setup):** Price stalls (`KER` dropping) while hourly funding spikes to extremes ($|F_{1\text{h}}| > 3$–8 bps/hr) and OI surges $\rightarrow$ high probability of a violent liquidation cascade in the opposite

direction (fatal for late Engine B breakouts, highly lucrative for Engine A contrarian entries once the flush begins).

- Should `funding` and `open_interest` delta ($\Delta \ln \mathrm{OI}_t$) be added as a 5th/6th emission dimension in `GaussianHMM`, or modeled as an explicit exogenous **Crowding / Squeeze Hazard Multiplier** in `RegimeController`? (Evaluate the curse of dimensionality and $N_{\text{eff}} \geq 40$ rare-state rejection constraints on a 1,000-bar 15m rolling training window).

2. **Continuous-Time O-U SDE under Stochastic Carry (F_t):**
   - Derive the exact adjustment to the Bertram optimal entry/exit thresholds (a^*, b^*) when holding the perpetual contract incurs a continuous carry rate $c_t = \text{side} \cdot F_{1\text{h}}(t)$ per hour.
   - How should the asymmetric shift of $(a^*_{\text{long}}, a^*_{\text{short}})$ be parameterized so that positive carry trades require a smaller price excursion (e.g., 1.30σ) while negative carry trades require a deeper price excursion (e.g., 1.75σ)?

3. Required Deliverables

1. **Theoretical & Empirical Trade-off Analysis:** 5D/6D HMM vs. 4D Price HMM + Exogenous Funding/OI Hazard Overlay (including impact on transition matrix stability and Baum-Welch convergence).
2. **Closed-Form Carry-Adjusted O-U Threshold Equations** for Long and Short entries/exits incorporating Hyperliquid's hourly funding schedule and mark-oracle premium.
3. **Production-Ready Rust Equations & Thresholds** for `BTC`, `ETH`, `SOL`, and `AVAX` (including Z-score normalization of funding and $\Delta\mathrm{OI}$ across assets with different baseline betas).

Recommendation: keep the HMM at 4 price-based emission dimensions and add funding plus open-interest dynamics as an exogenous "crowding/squeeze hazard" layer in `RegimeController`. With a 1,000-bar rolling window and your $N_{\text{eff}} \geq 40$ rare-state rule, a 5D/6D Gaussian HMM is likely to make the volatility-shock state less identifiable and destabilize Baum-Welch; funding/OI are better used to gate entries, scale size, and trigger contrarian squeeze logic. Hyperliquid's streamed `funding` is an hourly rate, while OI must be transformed before it is comparable across assets. [1] [2]

1. HMM integration trade-off

Dimensionality budget

For a D-dimensional full-covariance Gaussian emission, each state estimates D means and $D(D+1)/2$ covariance parameters.

Model	Emission params per state	Approx. params for 3 states + transitions	Assessment on 1,000 x 15m bars
4D price HMM	$4 + 10 = 14$	~46	Stable; matches your current design
5D: + funding	$5 + 15 = 20$	~64	Feasible only with shrinkage, winsorization, and careful scaling

Model	Emission params per state	Approx. params for 3 states + transitions	Assessment on 1,000 x 15m bars
6D: + funding, $\Delta \ln OI$	$6 + 21 = 27$	~85	High risk of overfitting and rare-state degeneracy

A 6D model needs roughly 85 free parameters from 1,000 observations. Your requirement that each state retain at least $N_{\text{eff}} \geq 40$ effective observations is especially binding for `State 2 = Volatility Shock`, which is already the rarest state. Adding two heavy-tailed, regime-dependent variables can cause the Baum-Welch responsibility mass for that state to collapse or oscillate. Ledoit-Wolf shrinkage helps, but it does not solve the underlying sample-size problem.[3]

Why funding/OI fit better exogenously

Funding and OI are **conditioning variables**, not necessarily latent-regime emissions:

- Funding is bounded, clipped, and mechanically related to the mark-oracle premium; it is not purely a market-state signal. Hyperliquid computes $F = \bar{P} + \text{clamp}(i - \bar{P}, -0.0005, 0.0005)$, where $i = 0.0001$ per 8 hours, and settles $1/8$ of the computed 8-hour rate each hour.[1]
- OI can rise in both healthy trend absorption and leveraged crowding; its interpretation depends on price, funding, and liquidation context.
- Your desired decision is asymmetric: a trend state with neutral funding should allow Donchian continuation, whereas the same trend state with extreme funding and rapid OI growth should suppress breakout entries or arm a squeeze playbook.

That is naturally a **hazard multiplier**, not another Gaussian emission dimension.

Recommended architecture

$$\text{Regime posterior } p(s_t \mid \mathcal{Y}_{1:t}) \quad \text{from 4D HMM}$$

$$H_t = h(F_t^z,\ \Delta \ln OI_t^z,\ \text{basis}_t^z,\ KER_t)$$

$$w_{\text{entry},t} = w_0 \cdot p(s_t = 1 \mid \mathcal{Y}_{1:t}) \cdot H_t$$

where $H_t \in [0, 1]$ reduces trend-following exposure when crowding is extreme, and can separately enable an Engine A contrarian squeeze state.

A practical hazard form is:

$$H_t = \exp\left(-\lambda_F \max(0, |F_t^z| - \tau_F) - \lambda_O \max(0, \Delta \ln OI_t^z - \tau_O) - \lambda_B \max(0, |\text{basis}_t^z| - \tau_B)\right)$$

with suggested starting values:

$$\lambda_F = 0.35, \quad \lambda_O = 0.25, \quad \lambda_B = 0.15$$

$$\tau_F = 2.0, \quad \tau_O = 2.0, \quad \tau_B = 1.5$$

Thus, a 3-sigma funding z-score with a 2.5-sigma OI surge can cut a trend breakout's size materially without corrupting the HMM's transition matrix.

If you still add emissions

Use a **5D model only**, with funding as the fifth feature, and only after these safeguards:

- Winsorize funding at $p_{0.5}$ and $p_{99.5}$, or model it with a robust/t distribution.
- Standardize each feature per asset using expanding or rolling robust statistics.
- Use diagonal or strongly shrunk covariance for the funding dimension.
- Reject a refit if any state's effective sample count falls below 40.
- Compare models by out-of-sample log-likelihood, transition-matrix persistence, state half-life, and realized squeeze hit-rate—not in-sample likelihood alone.

Do **not** put raw OI into the HMM. Use $\Delta \ln OI$ over a fixed horizon, ideally 1h or 4h, because OI is level-nonstationary and asset-scale dependent.

2. Hyperliquid funding and OI features

Hyperliquid's official formula is:

$$F_{8h} = \bar{P} + \text{clamp}(i - \bar{P},\ -0.0005,\ +0.0005)$$

$$i = 0.0001 \quad \text{per 8 hours}$$

$$F_{1h} = \frac{F_{8h}}{8}$$

The premium is derived from impact prices against the oracle:

$$P = \frac{\text{impact price difference}}{\text{oracle price}}$$

Longs pay shorts when the perp trades above oracle; shorts pay longs when it trades below. [4] [1]

For the bot, define signed carry from the perspective of the position:

$$c_t^{\text{long}} = -F_{1h,t}$$

$$c_t^{\text{short}} = +F_{1h,t}$$

A positive c_t means the position **receives** funding.

For open interest:

$$\Delta \ln OI_t = \ln OI_t - \ln OI_{t-\Delta}$$

Use $\Delta = 4$ bars on 15m data for a 1-hour OI impulse, and optionally a second feature at $\Delta = 16$ bars for a 4-hour accumulation measure.

Robust cross-asset normalization:

$$F^z_{i,t} = \frac{F_{i,t} - \text{median}(F_{i,t-L:t})}{\text{MAD}(F_{i,t-L:t})/0.6745}$$

$$\Delta OI^z_{i,t} = \frac{\Delta \ln OI_{i,t} - \text{median}(\Delta \ln OI_{i,t-L:t})}{\text{MAD}(\Delta \ln OI_{i,t-L:t})/0.6745}$$

Use $L = 672$ bars, i.e. seven days of 15m bars, and clip both z-scores to $[-5, 5]$. This avoids BTC's structurally lower funding beta contaminating SOL or AVAX signals.

3. Carry-adjusted O-U model

Let the mean-reverting spread or residual be:

$$dX_t = \kappa(\theta - X_t)\,dt + \sigma\,dW_t$$

where:

- κ is mean-reversion speed, per hour.
- θ is the long-run mean.
- σ is diffusion volatility, per $\sqrt{\text{hour}}$.
- X_t is standardized later into Z_t.

If holding one unit of the spread earns continuous carry c_t per hour, the trader's wealth process is:

$$dW_t^{\text{trade}} = dX_t + c_t\,dt$$

Substituting the O-U dynamics:

$$
\left[\kappa(\theta-X_t)+c_t\right]dt+\sigma\,dW_t
$$

This is equivalent to an O-U process with a carry-shifted mean:

$$\theta_t^{\text{eff}} = \theta + \frac{c_t}{\kappa}$$

because:

$$
\kappa(\theta^{eff}_t-X_t)
$$

\kappa(\theta-X_t)+c_t
]

Therefore, the exact first-order carry adjustment is a **parallel shift of the O-U center** by:

$$\Delta X_c = \frac{c_t}{\kappa}$$

In stationary standard-deviation units:

$$\sigma_{\text{stat}} = \frac{\sigma}{\sqrt{2\kappa}}$$

$$\Delta Z_c = \frac{c_t}{\kappa \sigma_{\text{stat}}} = \frac{c_t \sqrt{2}}{\sigma \sqrt{\kappa}}$$

Exact Bertram-style thresholds

Bertram's optimal symmetric solution has the property $b^* = -a^*$ around the process mean under the standardized O-U formulation. Let the base no-carry entry/exit half-width be a^*, expressed in stationary sigma units. [3]

For a long-spread trade:

$$Z_{\text{long}}^{\text{entry}} = -a^* + \Delta Z_c$$

$$Z_{\text{long}}^{\text{exit}} = +a^* + \Delta Z_c$$

For a short-spread trade:

$$Z_{\text{short}}^{\text{entry}} = +a^* + \Delta Z_c$$

$$Z_{\text{short}}^{\text{exit}} = -a^* + \Delta Z_c$$

Here $\Delta Z_c > 0$ means positive carry. Positive carry shifts the entire long entry/exit band upward: you enter after a smaller downside excursion and exit at a lower required recovery, exactly the economically desired behavior.

Practical asymmetric parameterization

The exact shift moves both thresholds. In production, you may prefer to preserve a fixed profit target and make only the **entry** carry-sensitive:

$$Z_{\text{long}}^{\text{entry}} = -(a^* - \gamma \Delta Z_c)$$

$$Z_{\text{short}}^{\text{entry}} = +(a^* - \gamma \Delta Z_c)$$

where $\gamma \in [0, 1]$ controls how aggressively carry relaxes the entry requirement.

To map directly to your 1.30σ versus 1.75σ example:

$$a_{\text{long}}^{\text{entry}} = \text{clamp}\left(a^* - \gamma \Delta Z_c,\ 1.30,\ 1.75\right)$$

$$a_{\text{short}}^{\text{entry}} = \text{clamp}\left(a^* - \gamma \Delta Z_c,\ 1.30,\ 1.75\right)$$

Use:

$$\gamma = 0.5$$

as a conservative starting value. Then positive carry reduces the required excursion, negative carry increases it, but the clamp prevents funding spikes from producing absurd entries during a liquidation cascade.

Exit thresholds

Keep exits risk-based rather than fully carry-shifted:

$$Z_{\text{long}}^{\text{exit}} = +\beta_{\text{exit}} a^*$$

$$Z_{\text{long}}^{\text{stop}} = -\beta_{\text{stop}} a^*$$

$$Z_{\text{short}}^{\text{exit}} = -\beta_{\text{exit}} a^*$$

$$Z_{\text{short}}^{\text{stop}} = +\beta_{\text{stop}} a^*$$

Suggested initial values:

$$\beta_{\text{exit}} = 0.35, \qquad \beta_{\text{stop}} = 2.25$$

This means: enter around 1.30σ–1.75σ against the mean, take profit at $0.35a^*$ through the mean, and stop at $2.25a^*$ beyond the entry.

4. Squeeze regime definition

Classify a squeeze setup only when all of the following occur:

$$p(s_t = 1 \mid \mathcal{Y}_{1:t}) > 0.60$$

$$|F_t^z| > 3.0$$

$$\Delta OI_t^z > 2.0$$

$$KER_{16,t} < KER_{16,\text{rolling median}} - 0.5 \cdot MAD(KER_{16})$$

$$|\text{basis}_t^z| > 1.5$$

Then:

- Disable or heavily reduce Engine B Donchian continuation entries in the funding direction.

- Arm Engine A only after a confirmed adverse displacement, e.g. a break of the prior 16-bar extreme plus a funding/OI persistence check.
- Do not fade merely because funding is high; require OI expansion and stalling trend efficiency to distinguish crowding from a genuine spot-led trend.

A useful persistence score is:

$$S_t = 0.45\,\mathbf{1}_{|F_t^z|>3} + 0.30\,\mathbf{1}_{\Delta OI_t^z>2} + 0.15\,\mathbf{1}_{KER_t<\tau_{KER}} + 0.10\,\mathbf{1}_{|\text{basis}_t^z|>1.5}$$

Enter squeeze mode when $S_t \geq 0.75$ for two consecutive 15m bars.

5. Production Rust equations

```
#[derive(Clone, Copy, Debug)]
pub struct MarketCtx {
    pub funding_1h: f64,      // Hyperliquid streamed hourly funding, decimal
    pub open_interest: f64,   // contracts or base units
    pub mark_px: f64,
    pub oracle_px: f64,
}

#[derive(Clone, Copy, Debug)]
pub struct RollingStats {
    pub funding_median: f64,
    pub funding_mad: f64,
    pub dlog_oi_median: f64,
    pub dlog_oi_mad: f64,
    pub basis_median: f64,
    pub basis_mad: f64,
}

pub struct OuParams {
    pub kappa: f64,           // per hour
    pub theta: f64,           // spread mean
    pub sigma: f64,           // diffusion vol, per sqrt(hour)
}

pub struct ThresholdConfig {
    pub base_a: f64,          // e.g. 1.55
    pub gamma: f64,           // e.g. 0.5
    pub min_entry_sigma: f64, // 1.30
    pub max_entry_sigma: f64, // 1.75
    pub exit_frac: f64,       // 0.35
    pub stop_frac: f64,       // 2.25
    pub max_abs_funding_z: f64,
}
```

Robust z-score

```
fn robust_z(x: f64, median: f64, mad: f64) -> f64 {
    let scale = (mad / 0.6745).max(1e-12);
    ((x - median) / scale).clamp(-5.0, 5.0)
}
```

Carry and threshold shift

```
fn position_carry(funding_1h: f64, side: Side) -> f64 {
    match side {
        Side::Long => -funding_1h,
        Side::Short => funding_1h,
    }
}

fn carry_z_shift(carry_1h: f64, ou: &OuParams) -> f64 {
    let sigma_stat = ou.sigma / (2.0 * ou.kappa).sqrt();
    let denom = (ou.kappa * sigma_stat).max(1e-12);
    (carry_1h / denom).clamp(-2.0, 2.0)
}
```

Entry thresholds

```
pub fn entry_sigma(
    side: Side,
    ctx: &MarketCtx,
    ou: &OuParams,
    cfg: &ThresholdConfig,
) -> f64 {
    let carry = position_carry(ctx.funding_1h, side);
    let dz = cfg.gamma * carry_z_shift(carry, ou);

    let raw = cfg.base_a - dz;

    // Positive carry => smaller required excursion.
    // Negative carry => deeper required excursion.
    raw.clamp(cfg.min_entry_sigma, cfg.max_entry_sigma)
}
```

Long thresholds

```
pub fn long_thresholds(
    ctx: &MarketCtx,
    ou: &OuParams,
    cfg: &ThresholdConfig,
) -> (f64, f64, f64) {
    let entry_sigma = entry_sigma(Side::Long, ctx, ou, cfg);

    let entry_z = -entry_sigma;
```

```
    let exit_z = cfg.exit_frac * cfg.base_a;
    let stop_z = -cfg.stop_frac * cfg.base_a;

    (entry_z, exit_z, stop_z)
}
```

Short thresholds

```
pub fn short_thresholds(
    ctx: &MarketCtx,
    ou: &OuParams,
    cfg: &ThresholdConfig,
) -> (f64, f64, f64) {
    let entry_sigma = entry_sigma(Side::Short, ctx, ou, cfg);

    let entry_z = entry_sigma;
    let exit_z = -cfg.exit_frac * cfg.base_a;
    let stop_z = cfg.stop_frac * cfg.base_a;

    (entry_z, exit_z, stop_z)
}
```

Crowding hazard

```
pub fn crowding_hazard(
    funding_z: f64,
    dlog_oi_z: f64,
    basis_z: f64,
    ker: f64,
    ker_median: f64,
) -> f64 {
    let f_term = 0.35 * (funding_z.abs() - 2.0).max(0.0);
    let oi_term = 0.25 * (dlog_oi_z - 2.0).max(0.0);
    let basis_term = 0.15 * (basis_z.abs() - 1.5).max(0.0);

    let ker_excess = (ker_median - ker).max(0.0);
    let ker_term = 0.10 * ker_excess;

    (-f_term - oi_term - basis_term - ker_term).exp().clamp(0.0, 1.0)
}
```

6. Initial per-asset thresholds

These are starting production values, not fitted constants. Calibrate κ, σ, and a^* per asset on rolling windows, then apply the same carry logic.

Asset	Base	Entry clamp	Funding z trigger	z trigger	Squeeze score
BTC	1.55	1.30σ–1.75σ	> 3.0	> 2.0	≥ 0.75
ETH	1.55	1.30σ–1.75σ	> 3.0	> 2.0	≥ 0.75

Asset	Base a^*	Entry clamp	Funding z trigger	$\Delta \ln OI$ z trigger	Squeeze score
SOL	1.60	1.30σ–1.80σ	> 3.0	> 2.25	≥ 0.75
AVAX	1.65	1.35σ–1.85σ	> 3.0	> 2.25	≥ 0.75

SOL and AVAX deserve wider OI thresholds because their OI and funding series tend to be noisier and more beta-sensitive than BTC or ETH. The robust z-score normalization already removes much of the level difference, but the trigger thresholds should still be asset-specific.

7. Validation plan

Before production:

1. **Backtest the 4D HMM unchanged**, then add the exogenous hazard overlay.
2. **Backtest a 5D funding HMM** only if the 4D-plus-overlay version shows a clear gap in squeeze avoidance or Sharpe.
3. Measure:
   - Trend-state transition persistence.
   - Fraction of `State 1` bars with $N_{\text{eff}} < 40$.
   - Baum-Welch convergence iterations and log-likelihood instability.
   - Late-breakout drawdown avoided by the hazard multiplier.
   - Engine A squeeze hit-rate, conditional on $S_t \geq 0.75$.
   - Carry-adjusted O-U entry distribution: verify positive-carry entries cluster nearer 1.30σ and negative-carry entries nearer 1.75σ.
4. Use purged, embargoed walk-forward splits; funding and OI effects are regime-dependent and leak easily through full-window normalization.

The cleanest production design is therefore: **4D Gaussian HMM for latent market state, funding/OI/basis for an exogenous squeeze hazard, and carry-shifted O-U thresholds for Engine A execution.** This preserves HMM stability while still making the bot sensitive to the leveraged-crowding conditions that price-only features cannot see. Hyperliquid's `ActiveAssetCtx` supplies the required real-time `funding`, `open_interest`, `mark_px`, and `oracle_px` fields needed for this pipeline. [2] [5]

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1. https://hyperliquid.gitbook.io/hyperliquid-docs/trading/funding
2. https://goldrush.dev/docs/api-reference/hyperliquid-info/web-data2
3. https://arxiv.org/pdf/2102.04160.pdf
4. https://hypermirror.io/hyperliquid/hyperliquid-funding-rates
5. https://www.dwellir.com/docs/hyperliquid/data-catalog/derived/market-context
6. https://rapidapi.com/liminal-labs-liminal-labs-default/api/hyperliquid-market-data-oi-funding-open-interest
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